

Reapti Architecture Dossier

Architectural Manifesto & System Overview

This document outlines the design foundations and specifications for the Reapti scientific design system. Powered by Material Design 3 Expressive and compile-time Panda CSS, each component enforces high performance, mathematical harmony, and strict accessibility.

Key Tenets

High-agency interface engineering with predictable tokens.

Zero layout thrashing through composable atomic primitives.

Deterministic typography scales and calibrated contrast ratios.

Full offline-first static asset generation.

Design Tokens & Page Layout

Standard ISO 216 Dimensions

The A4 page layout strictly honors standard ISO 216 proportions of 210 millimeters by 297 millimeters, yielding an exact aspect ratio of 1 to the square root of 2.

Visual Ergonomics

Elevated M3e cards provide spatial depth and focus.

Adaptive typography ensures legibility across digital viewports and high-DPI physical printouts.

Responsive scaling accommodates vertical continuous reading, horizontal carousels, and 3D page flip interactions.

Automated Static Pipeline

Playwright Headless PDF Generation

To guarantee fidelity across environments, PDF documents are generated during the build pipeline using headless Chromium driven by Playwright.

Pipeline Highlights

Zero runtime server dependencies for document downloads.

Automatic removal of interactive controls and floating action buttons during print rendering.

Bit-identical rendering between web previews and printed documents.

Direct static file serving through optimized content distribution.

